API Documentation

This document provides a full reference to all available API endpoints, including required parameters, authentication requirements, example requests, example responses, and relevant error codes. All responses are returned in JSON format.

1. JWT Authentication

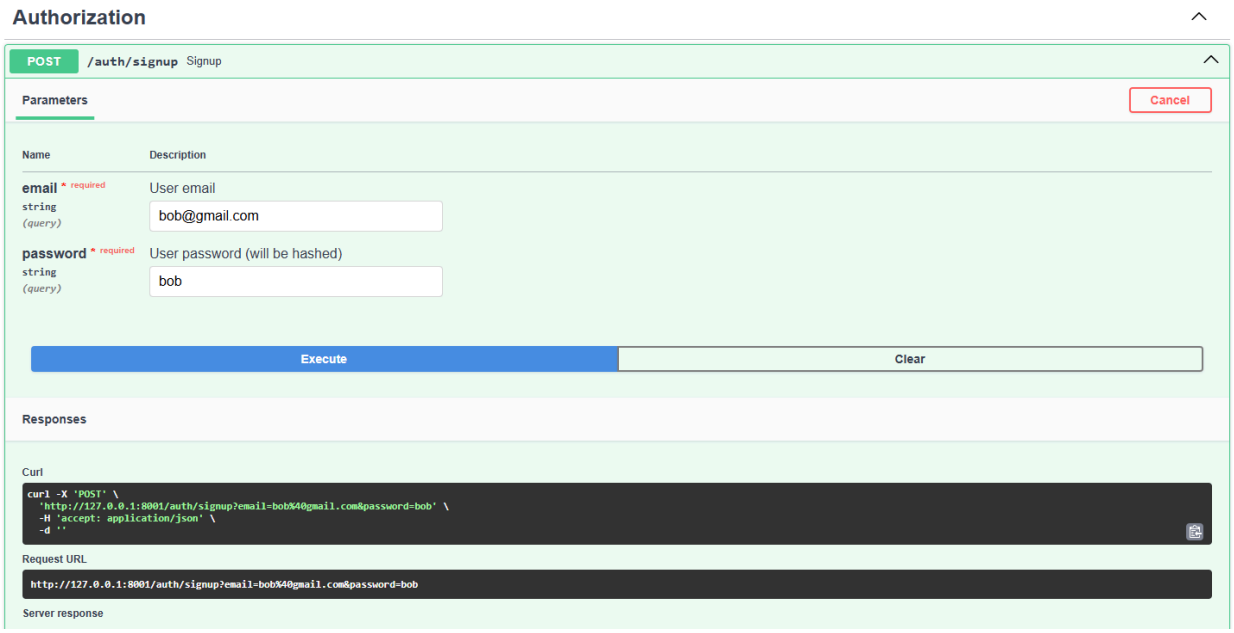
1.1 POST /auth/signup

Creates a new user account.

Query Parameters

Name	Type	Required	Description
email	string	yes	User email address
password	string	yes	Plaintext password

Example Request



Example Response (200)

Server response

Code	Details
200	Response body

```
{
  "id": 42,
  "email": "bob@gmail.com",
  "role": "user"
}
```

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Response headers

```
content-length: 47
content-type: application/json
date: Thu,19 Mar 2026 22:10:06 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value | Schema

```
"string"
```

422	Validation Error	*No links*

Media type

application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

Requests Producing Errors

Duplicate credentials

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8001/auth/signup?email=johnsmith%40gmail.com&password=johnsmith' \
  -H 'accept: application/json' \
  -d ''
```

Request URL

```
http://127.0.0.1:8001/auth/signup?email=johnsmith%40gmail.com&password=johnsmith
```

Server response

Code	Details
409 *Undocumented*	Error: Conflict

Response body

```
{
  "detail": "User with this email already exists."
}
```

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Response headers

```
content-length: 49
content-type: application/json
date: Thu,19 Mar 2026 22:07:20 GMT
server: uvicorn
```

1.2 POST /auth/login

Returns a JWT access token for authentication.

Example Request

POST `/auth/login` Login

Parameters | Cancel

Name	Description
email * required string *(query)*	User email johnsmith@gmail.com
password * required string *(query)*	User password johnsmith

Execute | Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8001/auth/login?email=johnsmith%40gmail.com&password=johnsmith' \
  -H 'accept: application/json' \
  -d ''
```

Request URL

```
http://127.0.0.1:8001/auth/login?email=johnsmith%40gmail.com&password=johnsmith
```

Example Response (200)

Server response

Code	Details
200	Response body


```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJqb2huc21pdGhAZ21haWwuY29tIiwicm9sZSI6InVzZXIiLCJ1c2VyX21kIjo0MSwiZXhwIjoxNzczOTYyMzA5fQ.kmNjvfDE3seQqCsfmgdVzkYV7go3r1DQM5rrPYN0RXg",
  "token_type": "bearer"
}
```


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Response headers

```
content-length: 220
content-type: application/json
date: Thu,19 Mar 2026 22:18:28 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type: application/json

Controls Accept header.

Example Value | Schema

```
{
  "access_token": "string",
  "token_type": "bearer"
}
```

422	Validation Error	*No links*

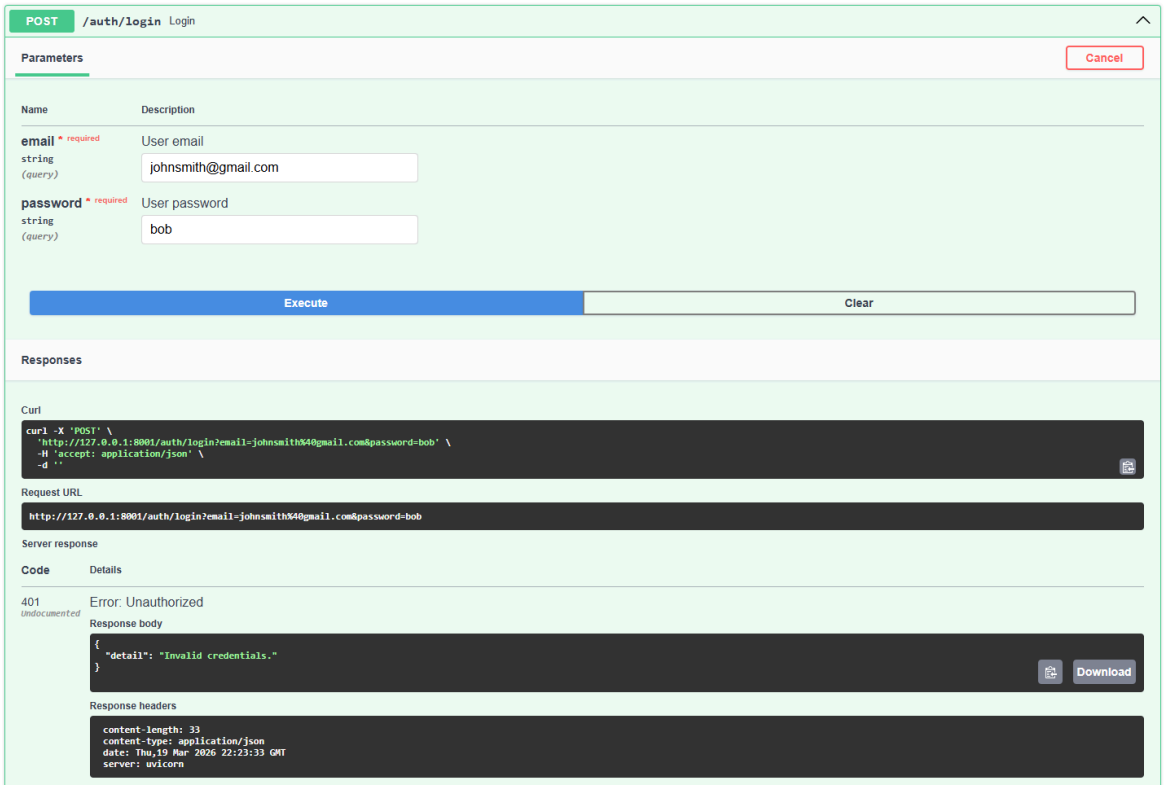
Media type: application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

Requests Producing Errors

Invalid credentials



Authentication Header Format

Authorization: Bearer <jwt-token>

2. Hotspot Endpoints

All hotspot endpoints are **public** (no authentication needed).

2.1 GET /hotspots/

Returns a list of all Wi-Fi hotspots.

Example Request

WiFi

GET `/hotspots/` List all WiFi hotspots

Returns the full list of enriched public WiFi hotspots, including core attributes, location coordinates, and any computed risk metadata.

Parameters | Cancel

No parameters

Execute | Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8080/hotspots/' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8080/hotspots/
```

Example Response

Server response

Code	Details
200	Response body


```
[
  {
    "wifi_id": "711e2c62-d6cc-466c-8e81-e20c3635960a",
    "name": "Akroyd Library",
    "city": "Calderdale",
    "status": null,
    "security_protection": null,
    "crime_12m_count": 510,
    "cyber_exposure_score": 50,
    "geom_geog": "0101000020E6100000D9EBDD1FEFD5FDBF1956F146E6DD4A40",
    "lat": 53.73359,
    "lon": -1.86473
  },
  {
    "wifi_id": "dcc198c4-ca28-4df3-801a-8db4e68e0212",
    "name": "Brighouse Library",
    "city": "Calderdale",
    "status": null,
    "security_protection": null,
    "crime_12m_count": 392,
    "cyber_exposure_score": 50,
    "geom_geog": "0101000020E6100000C190D5AD9E93FCBFBCAE5FB01BDA4A40",
    "lat": 53.70397,
    "lon": -1.78604
  },
  {
    "wifi_id": "33ef7c3f-a630-4056-93c5-e50a5e842711",
    "name": "Central Library and Archive",
```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Response headers

```
content-length: 161766
content-type: application/json
date: Thu,19 Mar 2026 22:33:14 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	List of all hotspots (enriched view) Media type: application/json Controls Accept header. Example Value | Schema	*No links*

```
"string"
```

2.2 GET /hotspots/search

Performs a case-insensitive match on hotspot names.

Example Request

GET /hotspots/search Search hotspots by name

Performs a case-insensitive match on hotspot names. Useful when the user types an area name, building name, or partial SSID.

Parameters Cancel

Name	Description
name * required string (query)	Partial hotspot name, e.g. `Civic` Civic

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8001/hotspots/search?name=Civic' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8001/hotspots/search?name=Civic
```

Example Response

Server response

Code	Details
200	Response body


```
[
  {
    "wifi_id": "29ef6382-6bbf-4f72-bff9-64ca6635d272",
    "name": "Leeds Civic Hall",
    "city": "Leeds",
    "status": "Live",
    "security_protection": "WPA2",
    "crime_12m_count": 3363,
    "cyber_exposure_score": 50,
    "geom_geog": "0101000020E610000078AE85DFA1C6F8BF64A99AC5B0E64A40",
    "lat": 53.80226965,
    "lon": -1.548404218
  },
  {
    "wifi_id": "ee3403a7-f834-4b0a-a543-a874e553ecac",
    "name": "Pudsey Civic Hall",
    "city": "Leeds",
    "status": "Live",
    "security_protection": "WPA2",
    "crime_12m_count": 150,
    "cyber_exposure_score": 50,
    "geom_geog": "0101000020E6100000084A2882BB8E6FABF542457985BE74A40",
    "lat": 53.80748276,
    "lon": -1.681327982
  }
]
```


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Response headers

```
content-length: 577
content-type: application/json
date: Thu,19 Mar 2026 22:36:08 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type: application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

Code	Description	Links
422	Validation Error	No links

Media type: application/json

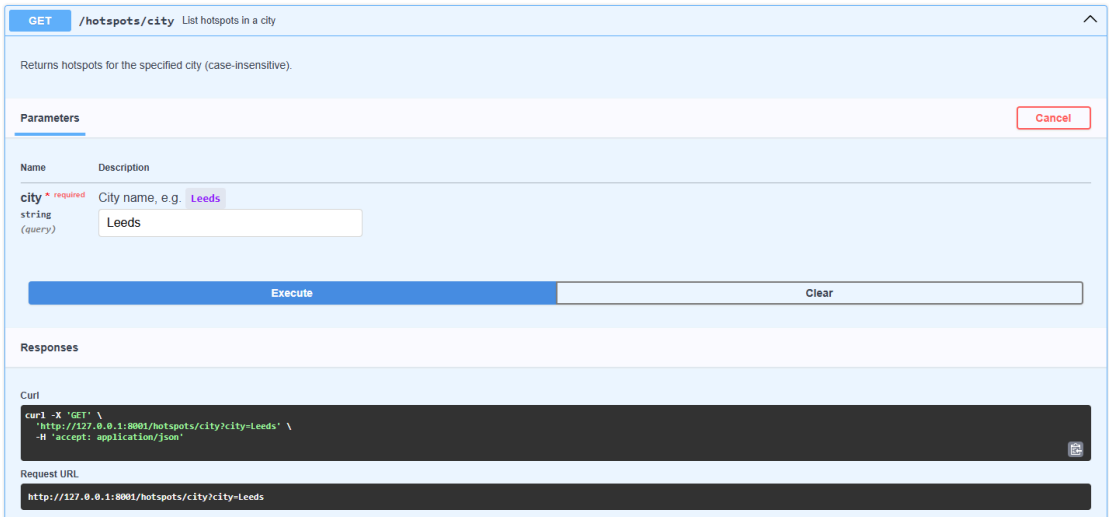
Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

2.3 GET /hotspots/city

Returns hotspots for the specified city.

Example Request



Example Response

Server response

Code	Details
200	Response body


```
[
  {
    "wifi_id": "2cae5c01-b3cd-4f06-8b75-1a17a83559a3",
    "name": "Abbey House Museum",
    "city": "Leeds",
    "status": "Live",
    "security_protection": "WPA2",
    "crime_12m_count": 111,
    "cyber_exposure_score": 50,
    "geom_geog": "0101000020E61000001A95F25FC2B6F9BFD13981E143E94A40",
    "lat": 53.82238406,
    "lon": -1.607118964
  },
  {
    "wifi_id": "01a641ec-6b1c-427d-bfda-2e59920912c6",
    "name": "Aberfield Gate Housing Office (BITMO)",
    "city": "Leeds",
    "status": "Live",
    "security_protection": "WPA2",
    "crime_12m_count": 431,
    "cyber_exposure_score": 50,
    "geom_geog": "0101000020E6100000EC0E0B0ED6FF8BF8F6335A7DCE04A40",
    "lat": 53.7567338,
    "lon": -1.5273262286
  },
  {
    "wifi_id": "598a62ff-5584-4ed1-b5f2-a3fb8c3dac21",
    "name": "Acorn Lodge Children's Unit",
    "city": "Leeds",
    "status": "Live",
```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Response headers

```
content-length: 68781
content-type: application/json
date: Thu,19 Mar 2026 22:46:49 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

422	Validation Error	*No links*

Media type

application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

2.4 GET /hotspots/nearest

Returns the nearest hotspot to a location.

Example Request

GET /hotspots/nearest Find nearest hotspot

Returns the closest WiFi hotspot to the given coordinates. Latitude and Longitude coordinates can be found from the previous endpoints.

Parameters

Cancel

Name	Description
lat * required number (query)	53.73359
lon * required number (query)	-1.86473

Execute | Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8001/hotspots/nearest?lat=53.73359&lon=-1.86473' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8001/hotspots/nearest?lat=53.73359&lon=-1.86473
```

Example Response

200

Response body


```
{
  "wifi_id": "711e2c62-d6cc-466c-8e81-e20c3635960a",
  "name": "Akroyd Library",
  "city": "Calderdale",
  "status": null,
  "security_protection": null,
  "crime_12m_count": 510,
  "cyber_exposure_score": 50,
  "geom_geog": "0101000020E6100000D9EBDD1FEFD5FDBF1956F146E6DD4A40",
  "lat": 53.73359,
  "lon": -1.86473,
  "dist": 0
}
```


Download

Response headers

```
content-length: 289
content-type: application/json
date: Thu,19 Mar 2026 22:58:53 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response Media type: application/json Controls Accept header. Example Value | Schema `"string"`	No links
422	Validation Error Media type: application/json Example Value | Schema	No links


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

2.5 GET /hotspots/nearest/knn

Returns the K nearest WiFi hotspots ordered by distance.

Example Request

GET /hotspots/nearest/knn Find K nearest hotspots

Returns the K nearest WiFi hotspots ordered by distance.

Parameters | Cancel

Name	Description
lat * required number (query)	53.73359
lon * required number (query)	-1.86473
k integer (query)	5

Execute | Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8001/hotspots/nearest/knn?lat=53.73359&lon=-1.86473&k=5' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8001/hotspots/nearest/knn?lat=53.73359&lon=-1.86473&k=5
```

Example Response

Code | Details

200

Response body


```
[
  {
    "wifi_id": "711e2c62-d6cc-466c-8e81-e20c3635960a",
    "name": "Akroyd Library",
    "city": "Calderdale",
    "status": null,
    "security_protection": null,
    "crime_12m_count": 518,
    "cyber_exposure_score": 50,
    "geom_geog": "0101000020E6100000D9EBDD1FEFD5FDBF1956F146E6DD4A40",
    "dist": 0
  },
  {
    "wifi_id": "4e77baba-9ce4-4ebd-9f85-41d54e6520df",
    "name": "Halifax Borough Market",
    "city": "Calderdale",
    "status": null,
    "security_protection": null,
    "crime_12m_count": 1829,
    "cyber_exposure_score": 50,
    "geom_geog": "0101000020E6100000C79DD2C1FABFFDBFB150689A77DC4A40",
    "dist": 1293.270223148158
  },
  {
    "wifi_id": "5c03bdcf-80c0-4138-b93c-ffb5b08b1e64",
    "name": "Victoria Theatre",
    "city": "Calderdale",
    "status": null,
    "security_protection": null,
    "crime_12m_count": 1879
```


Download

Response headers

```
content-length: 1379
content-type: application/json
date: Thu,19 Mar 2026 23:00:39 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type: application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

422	Validation Error	No links

Media type: application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

2.6 GET /hotspots/near

Returns hotspots within a distance (meters).

Example Request

GET /hotspots/near Find hotspots within radius

Returns all WiFi hotspots within the specified radius (meters).

Parameters

Cancel

Name	Description
lat * required number (query)	53.73359
lon * required number (query)	-1.86473
radius integer (query)	500

Execute | Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8001/hotspots/near?lat=53.73359&lon=-1.86473&radius=500' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8001/hotspots/near?lat=53.73359&lon=-1.86473&radius=500
```

Example Response

Server response

Code | Details

200

Response body


```
[
  {
    "wifi_id": "711e2c63-d6cc-466c-8e81-e20c363596 0a",
    "name": "Akroyd Library",
    "city": "Calderdale",
    "status": null,
    "security_protection": null,
    "crime_12m_count": 510,
    "cyber_exposure_score": 50,
    "geom_geog": "0101000020E6100000D9EBDD1FEFD5FDBF1956F146E6DD4A40",
    "lat": 53.73359,
    "lon": -1.86473,
    "dist": 0
  }
]
```


Download

Response headers

```
content-length: 291
content-type: application/json
date: Thu,19 Mar 2026 23:04:06 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

422	Validation Error	No links

Media type

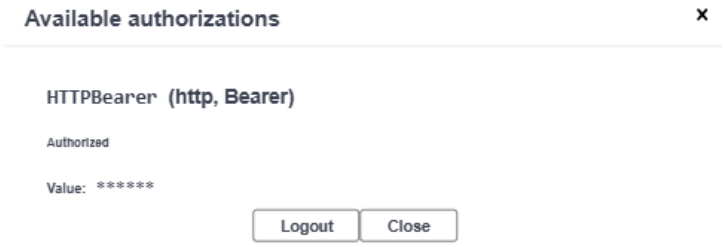
application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

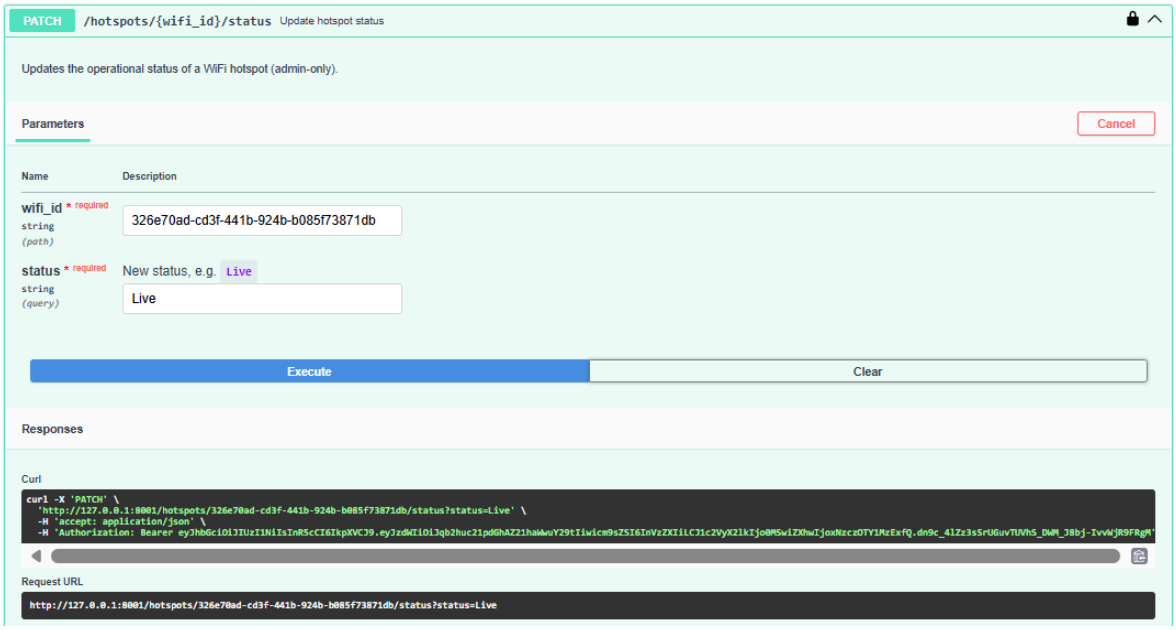
2.7 PATCH hotspots/{wifi_id}/status

Updates the operational status of a WiFi hotspot (admin-only) - **Authorization Needed**



Access token given at log in.

Example Request



Example Response

Server response

Code	Details
200	Response body

```
{
  "message": "Status updated"
}
```

Download

Response headers

```
content-length: 28
content-type: application/json
date: Thu,19 Mar 2026 23:10:36 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

422	Validation Error	*No links*

Media type

application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

Requests Producing Errors

Unauthorized

Responses

Curl

```
curl -X 'PATCH' \
  'http://127.0.0.1:8001/hotspots/326e70ad-cd3f-441b-924b-b085f73871db/status?status=Live' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer abvc'
```

Request URL

```
http://127.0.0.1:8001/hotspots/326e70ad-cd3f-441b-924b-b085f73871db/status?status=Live
```

Server response

Code	Details
401 *Undocumented*	Error: Unauthorized

Response body

```
{
  "detail": "Invalid or expired token"
}
```

Download

Response headers

```
content-length: 37
content-type: application/json
date: Thu,19 Mar 2026 23:14:40 GMT
server: uvicorn
```

2.7 PATCH hotspots/{wifi_id}/status

Updates the security mode of a WiFi hotspot (admin-only) - **Authorization Needed**

Example Request

PATCH /hotspots/{wifi_id}/security Update hotspot security mode

Updates the security mode of a WiFi hotspot (admin-only).

Parameters

Cancel

Name	Description
wifi_id * required string (path)	326e70ad-cd3f-441b-924b-b085f73871db
security_protection * required string (query)	One of: open , wpa2 , wpa3 open

Execute Clear

Responses

Curl

```
curl -X 'PATCH' \
  'http://127.0.0.1:8001/hotspots/326e70ad-cd3f-441b-924b-b085f73871db/security?security_protection=open' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJqb2huc21pdGhAZ21haWwuY29tIiwicm9sZSI6InVzZXIiLCJ1c2VyX2lkIjo0MSwiZXhwIjoxNzczOTY1MzExfQ.dn9c_4lZz3sSrUGuvTUVhS_DWM_J8bj-IvvWjR9FRgM'
```

Request URL

```
http://127.0.0.1:8001/hotspots/326e70ad-cd3f-441b-924b-b085f73871db/security?security_protection=open
```

Example Response

Server response

Code	Details
200	Response body

```
{
  "message": "Security updated"
}
```

Download

Response headers

```
content-length: 30
content-type: application/json
date: Thu,19 Mar 2026 23:16:23 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

Code	Description	Links
422	Validation Error	No links

Media type

application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

2.8 GET hotspots/{wifi_id}

Returns full hotspot details including metadata, security, crime exposure, risk score, and coordinates.

Example Request

PATCH /hotspots/{wifi_id}/security Update hotspot security mode

Updates the security mode of a WiFi hotspot (admin-only).

Parameters

Cancel

Name	Description
wifi_id * required string (path)	711e2c62-d6cc-466c-8e81-e20c3635960a
security_protection * required string (query)	One of: open , wpa2 , wpa3 open

Execute Clear

Responses

Curl

```
curl -X 'PATCH' \
  'http://127.0.0.1:8001/hotspots/711e2c62-d6cc-466c-8e81-e20c3635960a/security?security_protection=open' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJqb2huc21pdGhAZ21haWwuY29tIiwicm9sZSI6InVzZXIiLCJic2VyX2lkIjo0MSwiZXhwIjoxNzczOTY2MTkzfQ.gmc3O-bB-gl1bkc4zeLny0cVnqJaqiZHb366Fx7xmE'
```

Request URL

```
http://127.0.0.1:8001/hotspots/711e2c62-d6cc-466c-8e81-e20c3635960a/security?security_protection=open
```

Example Response

Server response

Code	Details
200	Response body

```
{
  "message": "Security updated"
}
```

Download

Response headers

```
content-length: 30
content-type: application/json
date: Thu,19 Mar 2026 23:23:47 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type: application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

Code	Description	Links
422	Validation Error	No links

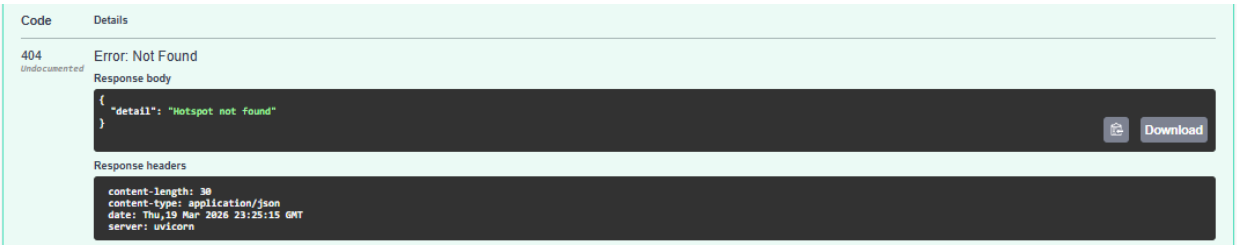
Media type: application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

Request Producing Error

Invalid {wifi_id}



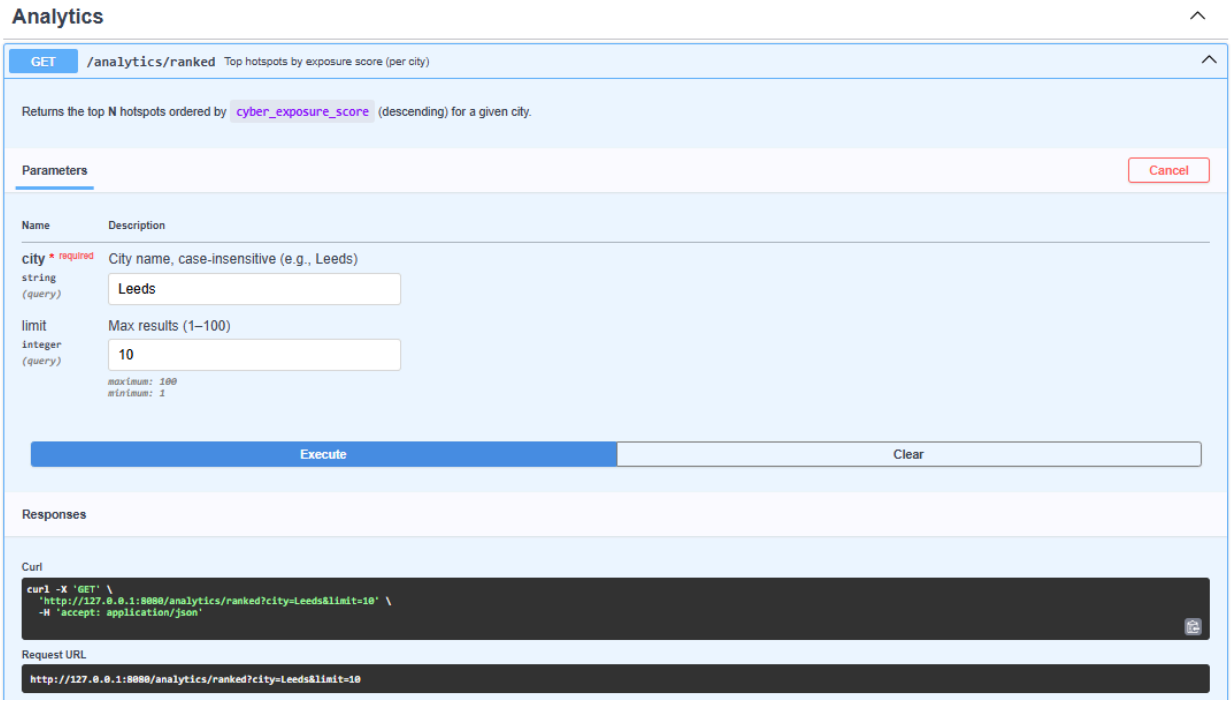
3. Analytics Endpoints

All analytics endpoints are **public** (no authentication needed).

3.1 GET /analytics/ranked

Returns hotspots ranked by cyber_exposure_score.

Example Request



Example Response

Server response

Code	Details
200	Response body


```
[
  {
    "wifi_id": "0133f11a-fce9-443e-a7d9-371f088f42b9",
    "name": "Lovell Park Hub",
    "postcode": "LS7 1DA",
    "city": "Leeds",
    "latitude": 53.80570239,
    "longitude": -1.53671205,
    "status": "Live",
    "security_protection": "WPA2",
    "cyber_exposure_score": 40,
    "risk_label": "caution",
    "crime_12m_count": 643,
    "incidents_count": 0,
    "last_incident_at": null,
    "geom_geog": "0101000020E6100000E7E7E15F5F96F8BFF5AD834121E74A40"
  },
  {
    "wifi_id": "01a641ec-6b1c-427d-bfda-2e59920912c6",
    "name": "Aberfield Gate Housing Office (BITMO)",
    "postcode": "LS10 3QH",
    "city": "Leeds",
    "latitude": 53.7567338,
    "longitude": -1.527326286,
    "status": "Live",
    "security_protection": "WPA2",
    "cyber_exposure_score": 40,
    "risk_label": "caution",
    "crime_12m_count": 433,
```


Download

Response headers

```
content-length: 3965
content-type: application/json
date: Thu,19 Mar 2026 23:59:23 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Top hotspots by risk score	*No links*

Media type: application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

422	Validation Error	*No links*

Media type: application/json

Example Value |


```
{
  "detail": [
    {
      "loc": [
        "query",
        "city"
      ],
      "msg": "field required",
      "type": "value_error.missing"
    }
  ]
}
```


3.2 GET /analytics/crime/{wifi_id}

Returns crime data for a specific hotspot.

Example Request

GET /analytics/crime/{wifi_id} Crime count near hotspot (12 months, 500 m)

Returns the pre-aggregated `crime_12m_count` for the specified hotspot.

Parameters

Cancel

Name	Description
wifi_id * required string *(path)*	711e2c62-d6cc-466c-8e81-e20c3635960a

Execute | Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/analytics/crime/711e2c62-d6cc-466c-8e81-e20c3635960a' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/analytics/crime/711e2c62-d6cc-466c-8e81-e20c3635960a
```

Example Response

Responses

Code	Description	Links
200	Crime count payload Media type application/json Controls Accept header. Example Value | Schema `"string"`	No links
404	Hotspot not found	No links
422	Validation Error Media type application/json Example Value | Schema	No links

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

Request Producing Error

Invalid {wifi_id}

Code	Details
404 *Undocumented*	Error: Not Found Response body `{ "detail": "Hotspot not found" }` Download Response headers

```
content-length: 30
content-type: application/json
date: Thu,19 Mar 2026 23:25:15 GMT
server: uvicorn
```

4. Incident Endpoints (JWT Protected)

These endpoints require a valid JWT.

4.1 POST /incidents/

Creates a new incident report.

Example Request

Incidents

POST `/incidents/` Create a new safety incident

Creates a **user-reported safety incident**, associated with a specific hotspot via its `wifi_id` and `bssid`. Incidents typically represent suspicious behaviour, device anomalies, or user-reported concerns. This endpoint **does not modify** any authoritative hotspot metadata, but contributes to user-facing risk intelligence.

Parameters Cancel

Name	Description
wifi_id * required string *(query)*	The hotspot WiFi ID 711e2c62-d6cc-466c-8e81-e20c3635960a
bssid * required string *(query)*	MAC/BSSID involved in incident 00:1A:2B:3C:4D:5E
description * required string *(query)*	Short description of the incident Pop-up ads *minLength: 3*

Execute | Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8080/incidents/?wifi_id=711e2c62-d6cc-466c-8e81-e20c3635960a&bssid=00%3A1A%3A2B%3A3C%3A4D%3A5E&description=Pop-up%20ads' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVC9.eyJzdWIiOiJqb2huc21pdGhAZ2ihaWwuY29tIiwicm9sZSI6InVzZXIiLCJic2VyX2lkIjo0MSwiZXhwIjoxNzczOTY4ODEzfQ.ErSi86VyfVqenLFPkW7pcOnYiIDWop-RjUAUzulMomY'
  -d ''
```

Request URL

```
http://127.0.0.1:8080/incidents/?wifi_id=711e2c62-d6cc-466c-8e81-e20c3635960a&bssid=00%3A1A%3A2B%3A3C%3A4D%3A5E&description=Pop-up%20ads
```

Example Response

Server response

Code	Details
201	Response body

```
{
  "id": 20,
  "wifi_id": "711e2c62-d6cc-466c-8e81-e20c3635960a",
  "bssid": "00:1A:2B:3C:4D:5E",
  "description": "Pop-up ads",
  "created_at": "2026-03-20T00:08:48.125530"
}
```

Download

Response headers

```
content-length: 155
content-type: application/json
date: Fri,20 Mar 2026 00:08:47 GMT
server: uvicorn
```

Responses

Code	Description	Links
201	Incident successfully created Media type: application/json Controls Accept header. Example Value | Schema `"string"`	*No links*
422	Validation error (missing or invalid fields)	*No links*

4.2 GET /incidents/{wifi_id}

Lists incidents for a hotspot.

Example Request

GET /incidents/{wifi_id} List incidents for a hotspot

Returns all incidents associated with a given **WiFi hotspot ID** (`wifi_id`), ordered by newest first. These reports typically originate from users or automated device warnings and can provide context for hotspot safety assessment.

Parameters

Cancel

Name	Description
wifi_id * required string *(path)*	711e2c62-d6cc-466c-8e81-e20c3635960a

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8080/incidents/711e2c62-d6cc-466c-8e81-e20c3635960a' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8080/incidents/711e2c62-d6cc-466c-8e81-e20c3635960a
```

Example Response

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8080/incidents/711e2c62-d6cc-466c-8e81-e20c3635960a' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8080/incidents/711e2c62-d6cc-466c-8e81-e20c3635960a
```

Server response

Code	Details
200	Response body


```
[
  {
    "id": 20,
    "wifi_id": "711e2c62-d6cc-466c-8e81-e20c3635960a",
    "bssid": "00:1A:2B:3C:4D:5E",
    "description": "Pop-up ads",
    "created_at": "2026-03-20T00:08:48.125530"
  }
]
```


Download

Response headers

```
content-length: 157
content-type: application/json
date: Fri,20 Mar 2026 00:15:18 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	List of incidents (may be empty)	*No links*
404	Hotspot has no recorded incidents	*No links*
422	Validation Error	*No links*

Media type

application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

Media type

application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

4.3 PATCH /incidents/{incident_id}

Updates an incident. {incident_id} can be retrieved from GET /incidents/{wifi_id}

Example Request

PATCH /incidents/{incident_id} Update an incident

Updates the description text of an existing incident. Retrieve incident IDs using GET /incidents/{wifi_id}.

Parameters

Cancel

Name	Description
incident_id * required integer (path)	20
description * required string (query)	New description for the incident Suspicious activity minLength: 3

Execute | Clear

Responses

Curl

```
curl -X 'PATCH' \
  'http://127.0.0.1:8080/incidents/20?description=Suspicious%20activity' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJqb2huc21pdGhAZ21haWwuY29tIiwicm9sZSI6InVzZXIiLCJ1c2VyX2lkIjo0MSwiZXhwIjoxNzczOTY5OTE0fQ.00tvxeVHuI89_DyWe5_2CK9vZy2bsehxKZv9Abmb1tk'
```

Request URL

```
http://127.0.0.1:8080/incidents/20?description=Suspicious%20activity
```

Example Response

Server response

Code	Details
200	Response body

```
{
  "id": 20,
  "wifi_id": "711e2c62-d6cc-466c-8e81-e20c3635960a",
  "bssid": "00:1A:2B:3C:4D:5E",
  "description": "Suspicious activity",
  "created_at": "2026-03-20T00:08:48.125530"
}
```

Download

Response headers

```
content-length: 164
content-type: application/json
date: Fri,20 Mar 2026 00:25:54 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Incident successfully updated	No links
404	Incident not found	No links
422	Validation error	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

4.4 DELETE /incidents/{incident_id}

Deletes an incident.

Example Request

DELETE /incidents/{incident_id} Delete an incident

Deletes the specified incident permanently. Retrieve incident IDs using GET /incidents/{wifi_id}`.

Parameters

Cancel

Name	Description
incident_id * required integer (path)	24

Execute | Clear

Responses

Curl

```
curl -X 'DELETE' \
  'http://127.0.0.1:8080/incidents/24' \
  -H 'accept: */*' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJqb2huc21pdGhAZ21haWwuY29tIiwicm9sZSI6InVzZXIiLCJic2VyX2lkIjo0MSwiZXhwIjoxNzczOTcwNjUyfQ.KVan5YfNG8ONIBFmwYozC655-OVIZtq1w8eEh09kqB4'
```

Request URL

```
http://127.0.0.1:8080/incidents/24
```

Example Response

Code	Details
204	Response headers

```
content-type: application/json
date: Fri,20 Mar 2026 00:40:09 GMT
server: uvicorn
```

Responses

Code	Description	Links
204	Incident successfully deleted	No links
404	Incident not found	No links
422	Validation Error	No links

Media type

application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

Requests Producing Error

Unauthorized

Server response

Code	Details
401 *Undocumented*	Error: Unauthorized

Response body

```
{
  "detail": "Not authenticated"
}
```

Download

Response headers

```
content-length: 30
content-type: application/json
date: Thu,19 Mar 2026 23:12:11 GMT
server: uvicorn
www-authenticate: Bearer
```

Incident Not Found

Curl

```
curl -X 'DELETE' \
  'http://127.0.0.1:8080/incidents/3' \
  -H 'accept: */*' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJqb2huc21pdGhAZ21haWwuY29tIiwicm9sZSI6InVzZXIiLCJ1c2VyX2lkIjo0MSwiZXhwIjoxNzczOTY5OTE0fQ.00tvxeVHuIB9_DyWe5_2CK9vZy2bsehxKZv9Abmb1tk'
```

Request URL

```
http://127.0.0.1:8080/incidents/3
```

Server response

Code	Details
404	Error: Not Found

Response body

```
{
  "detail": "Incident not found"
}
```

Download

Response headers

```
content-length: 31
content-type: application/json
date: Fri,20 Mar 2026 00:27:54 GMT
server: uvicorn
```

5. Assessment Endpoints

These endpoints analyse security, crime context, spoofing risk, and overall Wi-Fi safety.

All assessment endpoints are **public**.

5.1 GET /assessments/security

Returns the security protection level.

Example Request

Assessments

GET `/assessments/security` Security posture assessment (Open/WPA2/WPA3)

Returns the security mode of the identified hotspot (e.g., Open, WPA2, WPA3). This helps users understand the basic technical protection used by the network.

Parameters | Cancel

Name	Description
ssid * required string (query)	Akroyd Library
lat * required number (query)	53.70397
lon * required number (query)	-1.78604

Execute | Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8080/assessments/security?ssid=Akroyd%20Library&lat=53.70397&lon=-1.78604' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8080/assessments/security?ssid=Akroyd%20Library&lat=53.70397&lon=-1.78604
```

Example Response

Server response

Code	Details
200	Response body


```
{
  "ssid": "Akroyd Library",
  "wifi_id": "71e2c62-d6cc-466c-8e81-e20c3635960a",
  "security_protection": "open"
}
```


Download

Response headers

```
content-length: 103
content-type: application/json
date: Fri,20 Mar 2026 00:44:16 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type: application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

422	Validation Error	*No links*

Media type: application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

5.2 GET /assessments/crime

Returns crime incidents recorded within the hotspot's geospatial catchment.

Example Request

GET /assessments/crime Environmental crime risk assessment

Shows the number of recorded crime incidents within the hotspot's geospatial catchment area. Higher counts may indicate a riskier physical environment.

Parameters | Cancel

Name	Description
ssid * required string (query)	Akroyd Library
lat * required number (query)	53.70397
lon * required number (query)	-1.78604

Execute | Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8080/assessments/crime?ssid=Akroyd%20Library&lat=53.70397&lon=-1.78604' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8080/assessments/crime?ssid=Akroyd%20Library&lat=53.70397&lon=-1.78604
```

Example Response

Server response

Code	Details
200	Response body


```
{
  "ssid": "Akroyd Library",
  "wifi_id": "711e2c62-d6cc-466c-8e81-e20c3635960a",
  "crime_12m_count": 510
}
```


Download

Response headers

```
content-length: 96
content-type: application/json
date: Fri,20 Mar 2026 00:46:32 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type: application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

422	Validation Error	No links

Media type: application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

5.3 GET /assessments/incidents

Returns historical user-reported incidents.

Example Request

GET /assessments/incidents User-reported incident assessment

Lists recent user-reported cybersecurity or suspicious activity incidents associated with this hotspot. Reports may include phishing attempts, fake captive portals or strange network behaviour.

Parameters | Cancel

Name	Description
ssid * required string (query)	Akroyd Library
lat * required number (query)	53.73359
lon * required number (query)	-1.86473

Execute | Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8080/assessments/incidents?ssid=Akroyd%20Library&lat=53.73359&lon=-1.86473' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8080/assessments/incidents?ssid=Akroyd%20Library&lat=53.73359&lon=-1.86473
```

Example Response

Server response

Code	Details
200	Response body


```
{
  "ssid": "Akroyd Library",
  "wifi_id": "71ie2c62-d6cc-466c-8e81-e20c3635960a",
  "incidents": []
}
```


Download

Response headers

```
content-length: 89
content-type: application/json
date: Fri,20 Mar 2026 00:50:58 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type: application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

422	Validation Error	*No links*

Media type: application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

5.4 GET /assessments/ssid_risk

Detects whether an SSID is likely to be spoofed.

Example Request

GET /assessments/ssid_risk SSID spoofing likelihood assessment

Estimates whether the network name (SSID) is commonly used by malicious actors — for example names like 'Free WiFi', 'Airport-WiFi', 'Guest', or similar generic labels that are often spoofed.

Parameters

Cancel

Name	Description
ssid * required string (query)	Akroyd Library
lat * required number (query)	53.73359
lon * required number (query)	-1.86473

Execute | Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8080/assessments/ssid_risk?ssid=Akroyd%20Library&lat=53.73359&lon=-1.86473' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8080/assessments/ssid_risk?ssid=Akroyd%20Library&lat=53.73359&lon=-1.86473
```

Example Response

Server response

Code	Details
200	Response body

```
{
  "ssid": "Akroyd Library",
  "wifi_id": "711e2c62-d6cc-466c-8e81-e20c3635960a",
  "spoof_risk": "normal"
}
```

Download

Response headers

```
content-length: 96
content-type: application/json
date: Fri,20 Mar 2026 00:51:56 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

Code	Description	Links
422	Validation Error	No links

Media type

application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

5.5 GET /assessments/environment

Returns environmental/venue information and distance from the device.

Example Request

GET /assessments/environment Environmental and captive-portal risk assessment

Provides contextual information about the hotspot environment, including its operational status and distance from the provided location. Useful for distinguishing legitimate infrastructure from fake or suspicious networks.

Parameters

Cancel

Name	Description
ssid * required string (query)	Akroyd Library
lat * required number (query)	53.73359
lon * required number (query)	-1.86473

Execute | Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8080/assessments/environment?ssid=Akroyd%20Library&lat=53.73359&lon=-1.86473' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8080/assessments/environment?ssid=Akroyd%20Library&lat=53.73359&lon=-1.86473
```

Example Response

Server response

Code	Details
200	Response body


```
{
  "ssid": "Akroyd Library",
  "wifi_id": "711e2c62-d6cc-466c-8e81-e20c3635960a",
  "status": null,
  "distance_meters": 0
}
```


Download

Response headers

```
content-length: 110
content-type: application/json
date: Fri,20 Mar 2026 00:53:31 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type: application/json

Controls Accept header.

Example Value | Schema


```
"string"
```


422	Validation Error	No links

Media type: application/json

Example Value | Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

5.6 GET /assessments/safety

Final Cyber Exposure Score (0–100)

Includes crime, spoofing risk, incidents, and security mode.

Example Request

GET /assessments/safety General Wi-Fi Safety Assessment

Returns a complete Wi-Fi safety verdict based on SSID + location, using the **cyber_exposure_score stored in the database**. Combines security, crime, SSID risk, environmental factors and final recommendations.

🟢 **0–30: Low Risk** — Strong security, low crime, no suspicious activity. 🟡 **31–60: Medium Risk** — Some concerns such as weaker security or minor incidents. 🔴 **61–100: High Risk** — Open networks, suspicious incidents, high crime areas.

Parameters

Cancel

Name	Description
ssid * required string (query)	SSID of the Wi-Fi network Akroyd Library
lat * required number (query)	Latitude of the device 53.73359
lon * required number (query)	Longitude of the device -1.86473

Execute | Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8003/assessments/safety?ssid=Akroyd%20Library&lat=53.73359&lon=-1.86473' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8003/assessments/safety?ssid=Akroyd%20Library&lat=53.73359&lon=-1.86473
```

Example Response

Responses

Code	Description	Links
200	Composite safety assessment Media type: application/json Controls Accept header. Example Value | Schema `"string"`	No links
422	Validation Error Media type: application/json Example Value | Schema	No links

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

Requests Producing Error

Insufficient Information on Network

Code Details

404 Undocumented Error: Not Found

Response body

```
{
  "detail": "Insufficient information for this network"
}
```

Download

Response headers

```
content-length: 54
content-type: application/json
date: Fri,20 Mar 2026 01:48:05 GMT
server: uvicorn
```

6. Internal Endpoints (API Key)

These endpoints require a valid API Key.

6.1 GET /internal/stats

Returns internal statistics.

Example Request

Internal

GET /internal/stats Internal Stats

Parameters Cancel

Name	Description
x-api-key * required string (header)	abc123supersecretkey987

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8003/internal/stats' \
  -H 'accept: application/json' \
  -H 'x-api-key: abc123supersecretkey987'
```

Request URL

```
http://127.0.0.1:8003/internal/stats
```

Example Response

Server response

Code	Details
200	Response body

```
{
  "ok": true,
  "database": "connected",
  "hotspots": 543,
  "incidents": 0,
  "crime_records": 2904198,
  "uptime_seconds": 202
}
```

Download

Response headers

```
content-length: 108
content-type: application/json
date: Fri,20 Mar 2026 01:58:07 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

422	Validation Error	*No links*

Media type

application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

6.2 GET /internal/version

Returns, API version, PostgreSQL version, PostGIS version.

Example Request

GET /internal/version Version Info

Parameters

Cancel

Name	Description
x-api-key * required string (header)	abc123supersecretkey987

Execute | Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8003/internal/version' \
  -H 'accept: application/json' \
  -H 'x-api-key: abc123supersecretkey987'
```

Request URL

```
http://127.0.0.1:8003/internal/version
```

Example Response

Server response

Code	Details
200	Response body


```
{
  "api_version": "unknown",
  "postgres_version": "PostgreSQL 18.3 on x86_64-windows, compiled by msvc-19.44.35223, 64-bit",
  "postgis_version": "POSTGIS=\"3.6.1 3.6.1\" [EXTENSION] PGSQL=\"180\" GEOS=\"3.14.1-CAPI-1.20.5\" PROJ=\"8.2.1 NETWORK_ENABLED=OFF URL_ENDPOINT=https://cdn.proj.org USER_WRITABLE_DIRECTORY=
C:\\WINDOWS\\ServiceProfiles\\NetworkService\\AppData\\Local/proj DATABASE_PATH=C:\\Program Files\\PostgreSQL\\18\\share\\contrib\\postgis-3.6\\proj\\proj.db\" (compiled against PROJ 8.2.1) LIBX
ML=\"2.12.5\" LIBJSON=\"0.12\" LIBPROTOBUF=\"1.2.1\" WAGYU=\"0.5.0 (Internal)\" TOPOLOGY"
}
```


Download

Response headers

```
content-length: 590
content-type: application/json
date: Fri,20 Mar 2026 01:58:40 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

422	Validation Error	*No links*

Media type

application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

Request Producing Error

Code	Details
401 *Undocumented*	Error: Unauthorized

Response body

```
{
  "detail": "Invalid API Key"
}
```

Download

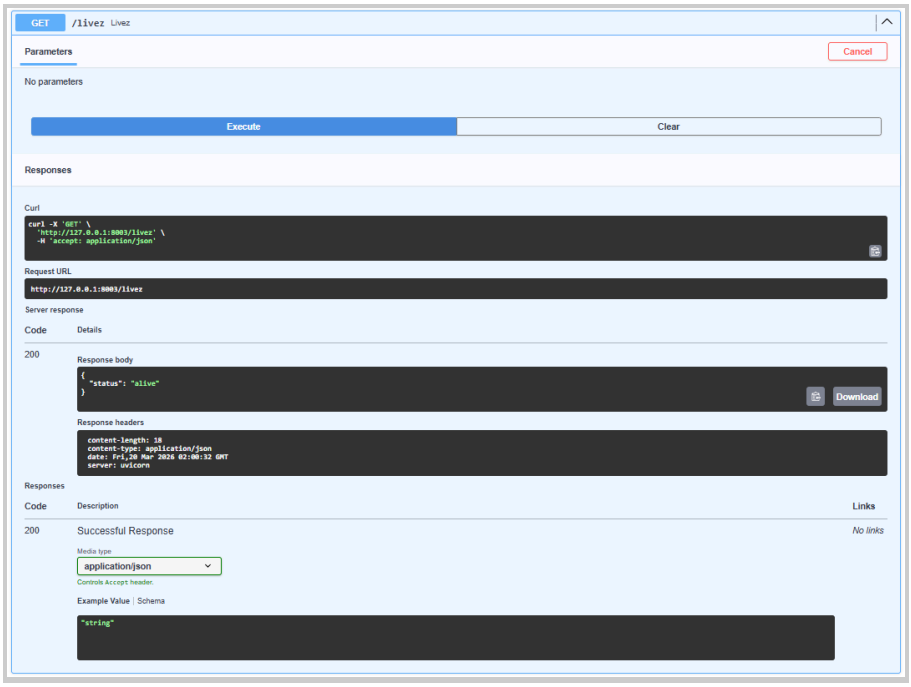
Response headers

```
content-length: 28
content-type: application/json
date: Fri,20 Mar 2026 01:56:16 GMT
server: uvicorn
```

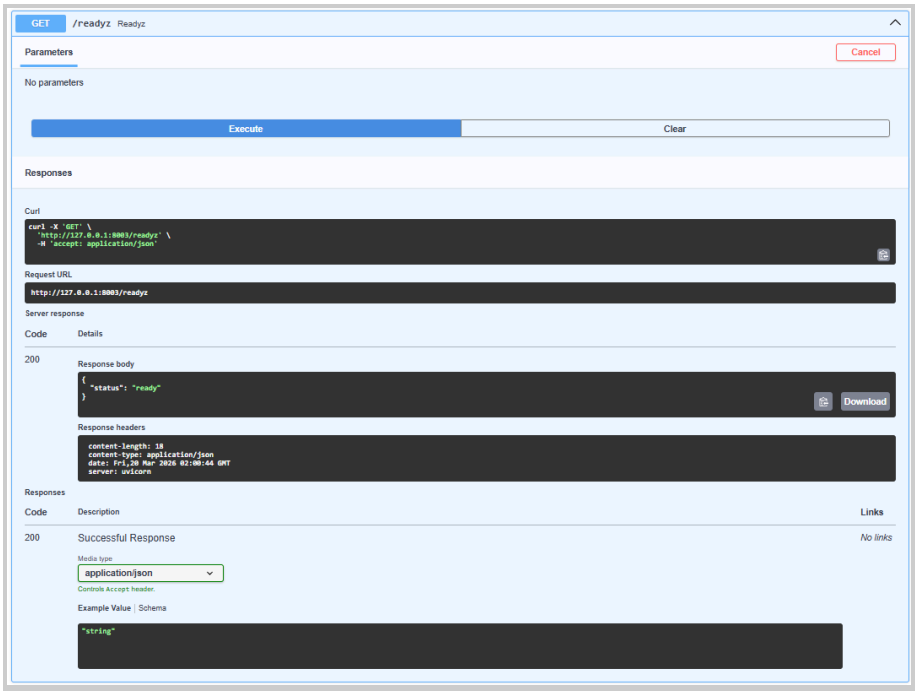
7. Default Endpoints

Returns a simple status indicator confirming that the API is running. Useful for external services, uptime monitoring, and CI/CD scripts that only need to know whether the application process is alive.

7.1 GET /livez



7.2 GET /readyz



7.3 GET /healthz

GET /healthz Healthz

Parameters

Cancel

No parameters

Execute | Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8003/healthz' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8003/healthz
```

Server response

Code	Details
200	Response body

```
{
  "status": "ok"
}
```

Download

Response headers

```
content-length: 15
content-type: application/json
date: Fri,20 Mar 2026 02:04:15 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

7. Schemas

Schemas

HTTPValidationError Collapse all object

- detail Collapse all array<object>
  - Items Collapse all object
    - loc* Collapse all array<(string | integer)>
      - Items Collapse all (string | integer)
        - Any of Collapse all (string | integer)
          - #0 string
          - #1 integer
    - msg* string
    - type* string
    - input any
    - ctx object

TokenResponse Collapse all object

- access_token* string
- token_type Collapse all string
  - Default="bearer"

ValidationError Collapse all object

- loc* Collapse all array<(string | integer)>
  - Items Collapse all (string | integer)
    - Any of Collapse all (string | integer)
      - #0 string
      - #1 integer
- msg* string
- type* string
- input any
- ctx object

8. Parameters

8.1 Authentication

POST /auth/signup

Parameter	Location	Type	Description
email	query	string	User email
password	query	string	Plain user password

POST /auth/login

Parameter	Location	Type	Description
email	query	string	User email
password	query	string	User password

8.2 WiFi Hotspots

GET /hotspots/

No parameters

GET /hotspots/search

Parameter	Location	Type	Description
name	query	string	Partial hotspot name

GET /hotspots/city

Parameter	Location	Type
city	query	string

GET /hotspots/nearest

Parameter	Location	Type
lat	query	float
lon	query	float

GET /hotspots/nearest/knn

Parameter	Location	Type	Default
lat	query	float	—
lon	query	float	—
k	query	int	5

GET /hotspots/near

Parameter	Location	Type	Default
lat	query	float	—
lon	query	float	—
radius	query	int	500

PATCH /hotspots/{wifi_id}/status

Parameter	Location	Type
wifi_id	path	string
status	query	string

PATCH /hotspots/{wifi_id}/security

Parameter	Location	Type
wifi_id	path	string
security_protection	query	string

8.3 Analytics

GET /analytics/ranked

Parameter	Location	Type	Default	Description
city	query	string	—	City name
limit	query	int	10	Number of results (1–100)

GET /analytics/crime/{wifi_id}

Parameter	Location	Type
wifi_id	path	string

8.4 Incidents

POST /incidents/

Parameter	Location	Type	Description
wifi_id	query	string	Hotspot ID
bssid	query	string	BSSID involved in incident
description	query	string	Incident description

GET /incidents/{wifi_id}

Parameter	Location	Type
wifi_id	path	string

PATCH /incidents/{incident_id}

Parameter	Location	Type
incident_id	path	int
description	query	string

DELETE /incidents/{incident_id}

Parameter	Location	Type
incident_id	path	int

8.5 Assessments

GET /assessments/security

Parameter	Location	Type
ssid	query	string
lat	query	float
lon	query	float

GET /assessments/crime

Parameter	Location	Type
ssid	query	string
lat	query	float
lon	query	float

GET /assessments/incidents

Parameter	Location	Type
ssid	query	string
lat	query	float
lon	query	float

GET /assessments/ssid_risk

Parameter	Location	Type
ssid	query	string
lat	query	float
lon	query	float

GET /assessments/environment

Parameter	Location	Type
ssid	query	string
lat	query	float
lon	query	float

GET /assessments/safety

Parameter	Location	Type
ssid	query	string
lat	query	float
lon	query	float

8.6 Internal Endpoints

GET /internal/stats

No parameters (API key required in header).

Header	Type	Description
X-API-Key	string	Required API key for internal endpoints

GET /internal/version

No parameters (API key required in header).

Header	Type	Description
X-API-Key	string	Required API key for internal endpoints

8.7 Default System Endpoints

GET /healthz

No parameters

Parameter	Location	Type	Description
—	—	—	No parameters required

GET /livez

No parameters

Parameter	Location	Type	Description
—	—	—	No parameters required

GET /readyz

No parameters

Parameter	Location	Type	Description
—	—	—	No parameters required

9. Error Codes

Status Code	Error Meaning	When It Occurs (Endpoint Category)
400 Bad Request	Invalid or unsupported parameter value	Hotspots (invalid status/security), Assessments (invalid inputs), Analytics (invalid limit)
401 Unauthorized	Missing or invalid JWT token	Incidents (POST/PATCH/DELETE), Hotspot admin updates
403 Forbidden	Missing or invalid API key	Internal endpoints /internal/*
404 Not Found	Resource not found	Hotspots (wifi_id), Incidents (incident_id, empty list), Analytics crime lookup, Assessments (SSID could not be resolved)
409 Conflict	Resource already exists	Authentication (signup with existing email)
422 Unprocessable Entity	Validation failed (wrong types, missing required fields)	All endpoints with typed query params (lat/lon/ssid/city/name/description)
500 Internal Server Error	Unexpected server error or database failure	Any endpoint (uncaught exception)
503 Service Unavailable	Service dependency unavailable	/readyz when database is not reachable, internal diagnostics failure